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ENROLLMENT NUMBER:-S25MCAG0010

LAB_ASSIGNMENT-10

Q1. Insert a new customer and open an account.

```
mysql> SELECT * FROM customers;
+---------+--------+---------+------------+
| Cust_id | Cname  | City    | Phone      |
+---------+--------+---------+------------+
|     101 | Raj    | Delhi   | 9876543210 |
|     102 | Simran | Mumbai  | 9123456780 |
|     103 | Amit   | Kolkata | 9988776655 |
|     104 | Ankit  | Delhi   | 9999999999 |
+---------+--------+---------+------------+
4 rows in set (0.00 sec)

mysql> INSERT INTO Accounts VALUES (2004, 104, 'Savings', 12000);
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM accounts;
+--------+---------+----------+----------+
| Acc_no | Cust_id | Acc_type | Balance  |
+--------+---------+----------+----------+
|   2001 |     101 | Savings  | 10000.00 |
|   2002 |     102 | Current  | 15000.00 |
|   2003 |     103 | Savings  |  8000.00 |
|   2004 |     104 | Savings  | 12000.00 |
+--------+---------+----------+----------+
4 rows in set (0.00 sec)
```

Q2. Deposit ₹2000 into account 2001.

```
mysql> UPDATE Accounts
    -> SET Balance = Balance + 2000
    -> WHERE Acc_no = 2001;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> INSERT INTO Transactions VALUES (9003, 2001, 'Deposit', 2000, CURRENT
_DATE);
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM transactions;
+------+--------+----------+---------+------------+
| T_id | Acc_no | T_type   | Amount  | T_date     |
+------+--------+----------+---------+------------+
| 9001 |   2001 | Deposit  | 5000.00 | 2023-10-01 |
| 9002 |   2002 | Withdraw | 2000.00 | 2023-10-02 |
| 9003 |   2001 | Deposit  | 2000.00 | 2026-04-20 |
+------+--------+----------+---------+------------+
3 rows in set (0.00 sec)
```

Q3. Withdraw ₹1000 from account 2001 and rollback.

```
mysql> UPDATE Accounts
    -> SET Balance = Balance - 1000
    -> WHERE Acc_no = 2001;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from accounts;
+--------+---------+----------+----------+
| Acc_no | Cust_id | Acc_type | Balance  |
+--------+---------+----------+----------+
|   2001 |     101 | Savings  | 17000.00 |
|   2002 |     102 | Current  | 15000.00 |
|   2003 |     103 | Savings  |  8000.00 |
|   2004 |     104 | Savings  | 12000.00 |
+--------+---------+----------+----------+
4 rows in set (0.00 sec)
```

Q4. Transfer ₹3000 from Acc_no 2001 to Acc_no 2002.

```
mysql> UPDATE Accounts
    -> SET Balance = Balance - 3000
    -> WHERE Acc_no = 2001;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE Accounts
    -> SET Balance = Balance + 3000
    -> WHERE Acc_no = 2002;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

Q5. Delete a customer (Cust_id=101), create a savepoint, and rollback.

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> SAVEPOINT sp1;
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELETE FROM Customers WHERE Cust_id = 101;
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> ROLLBACK TO sp1;
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

Q6. Update account type of Acc_no=2002 to “Savings” and commit.

```
mysql> UPDATE Accounts
    -> SET Acc_type = 'Savings'
    -> WHERE Acc_no = 2002;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)
```

Q7. Insert two new customers with savepoints, rollback second only.

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> -- Use new unique IDs
mysql> INSERT INTO Customers VALUES (201, 'Riya', 'Pune', '8888888888');
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> SAVEPOINT sp1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO Customers VALUES (202, 'Karan', 'Delhi', '7777777777');
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> ROLLBACK TO sp1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
```

Q8. Deduct ₹500 from all savings accounts and rollback.

```
mysql> UPDATE Accounts
    -> SET Balance = Balance - 500
    -> WHERE Acc_type = 'Savings';
Query OK, 4 rows affected (0.01 sec)
Rows matched: 4  Changed: 4  Warnings: 0

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql> |
```

Q9. Insert a new transaction record without committing, then rollback.

```
mysql> INSERT INTO Transactions VALUES (9004, 2001, 'Deposit', 1000, current
_date);
Query OK, 1 row affected (0.01 sec)

mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql> |
```

Q10. Delete all accounts with balance < 5000, then rollback.

```
mysql> DELETE FROM Transactions
    -> WHERE Acc_no IN (
    ->     SELECT Acc_no FROM Accounts WHERE Balance < 5000
    -> );
Query OK, 3 rows affected (0.01 sec)

mysql>
mysql> DELETE FROM Accounts
    -> WHERE Balance < 5000;
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql> |
```